



CREATE Garden Shears: Garden Reliably, Inclusively, and Painlessly (GRIP)



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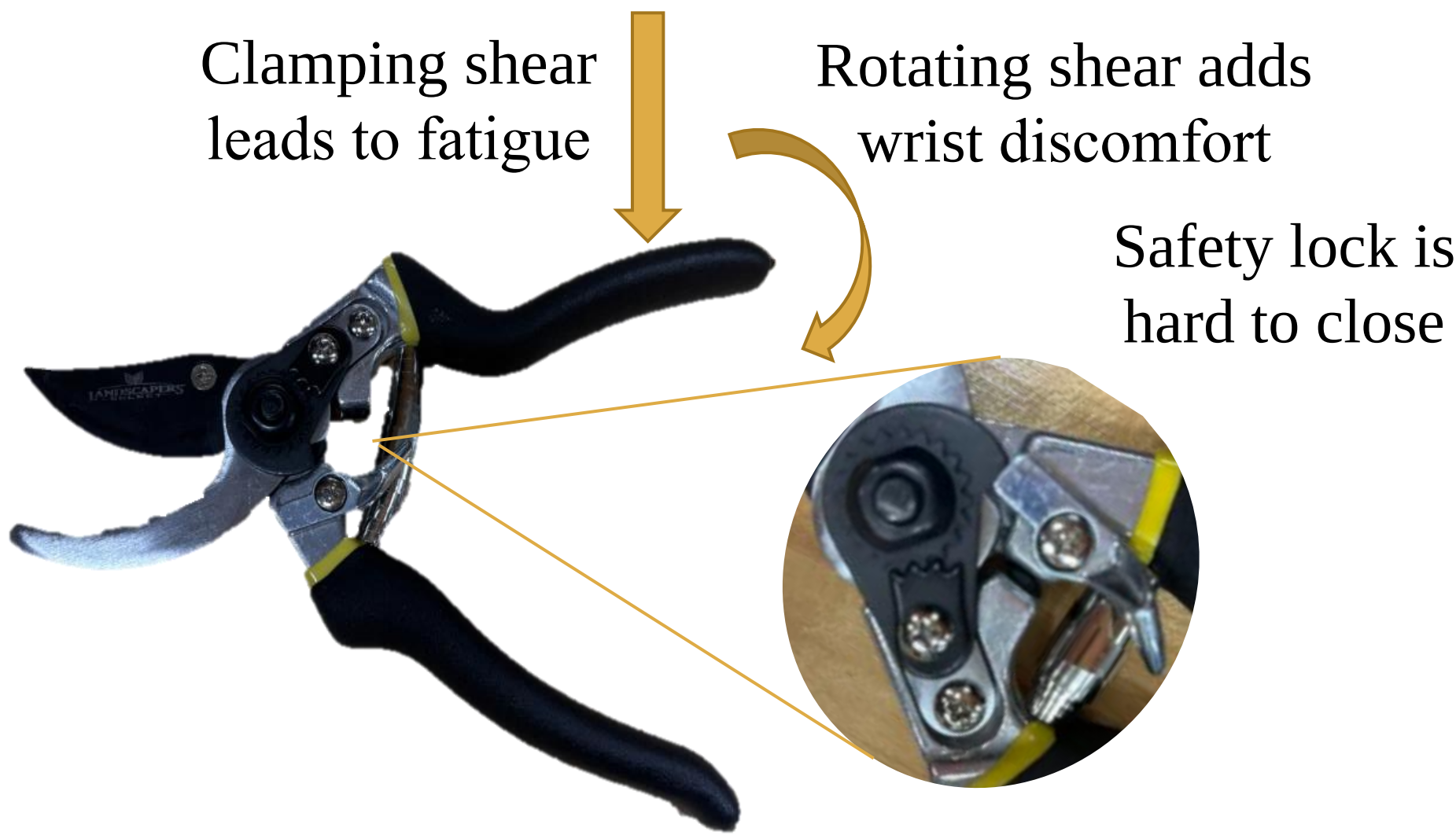
Project Overview

Harvesting garden vegetables with the use of garden tools, such as clippers and shears, can be difficult for people who suffer from disabilities that impact hand mobility and dexterity. These clippers, especially with rust, can be difficult to lock, and the repeated motion of shearing and rotating causes fatigue for users in their hands.

User and Client

Client: CP Unlimited, one of New York State Industries for the Disabled's (NYSID) partner organizations, hosts Smile Farms
User: Disabled employees at Smile Farms

Issues with Current Shears



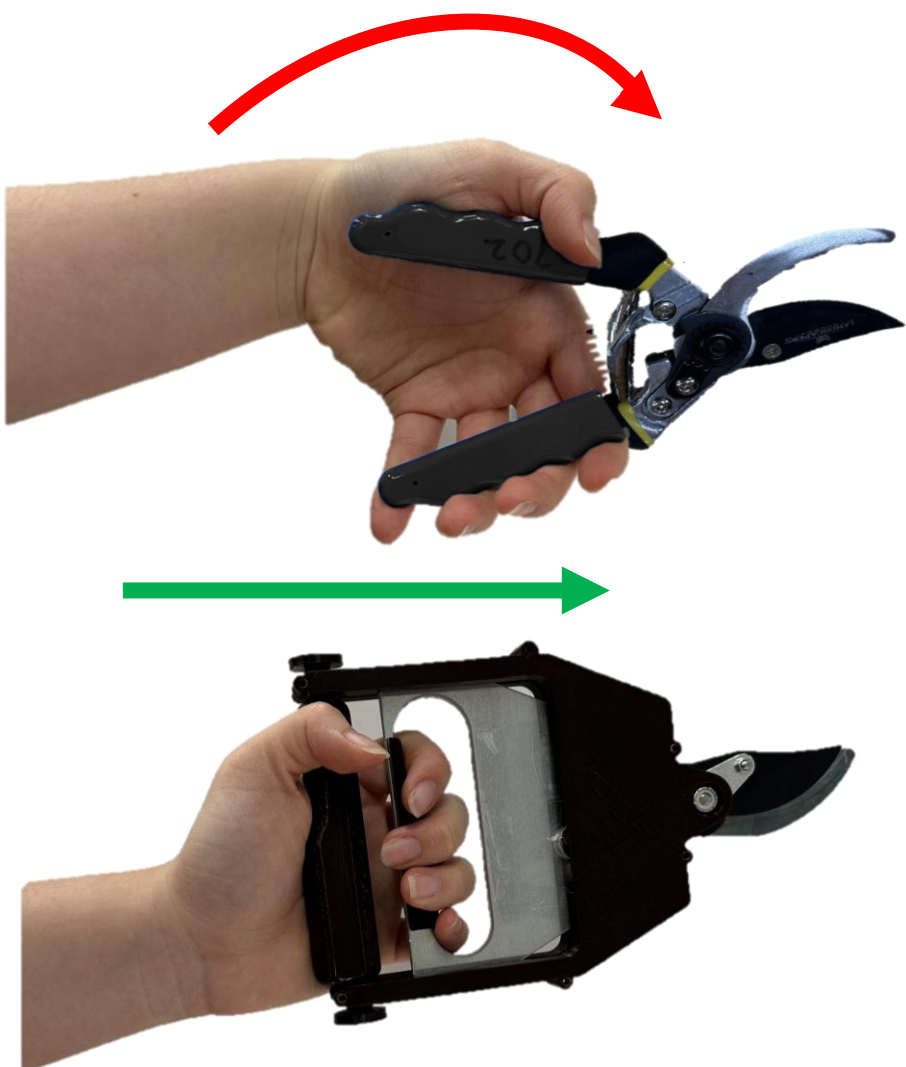
Design Choices

Comfort

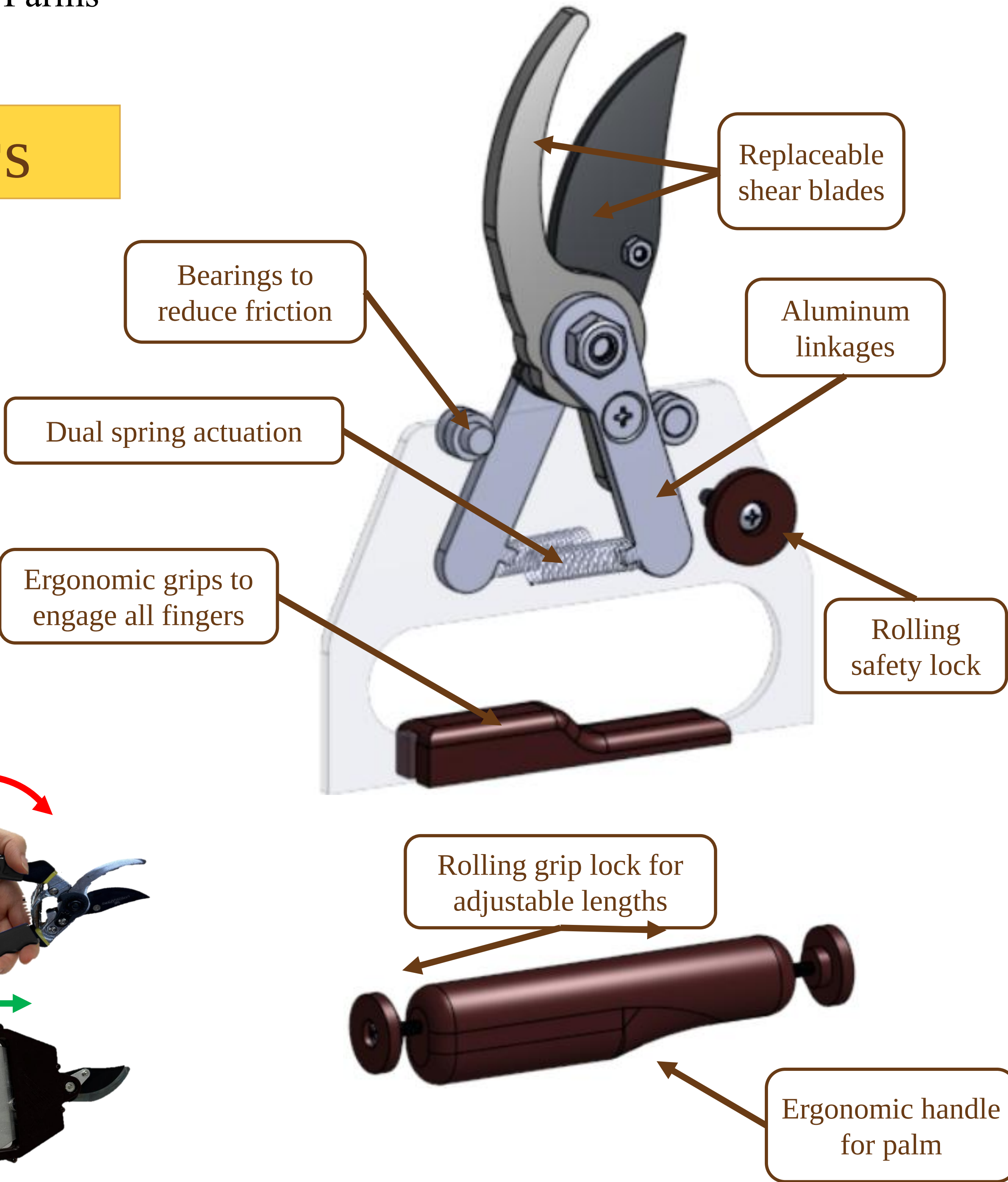
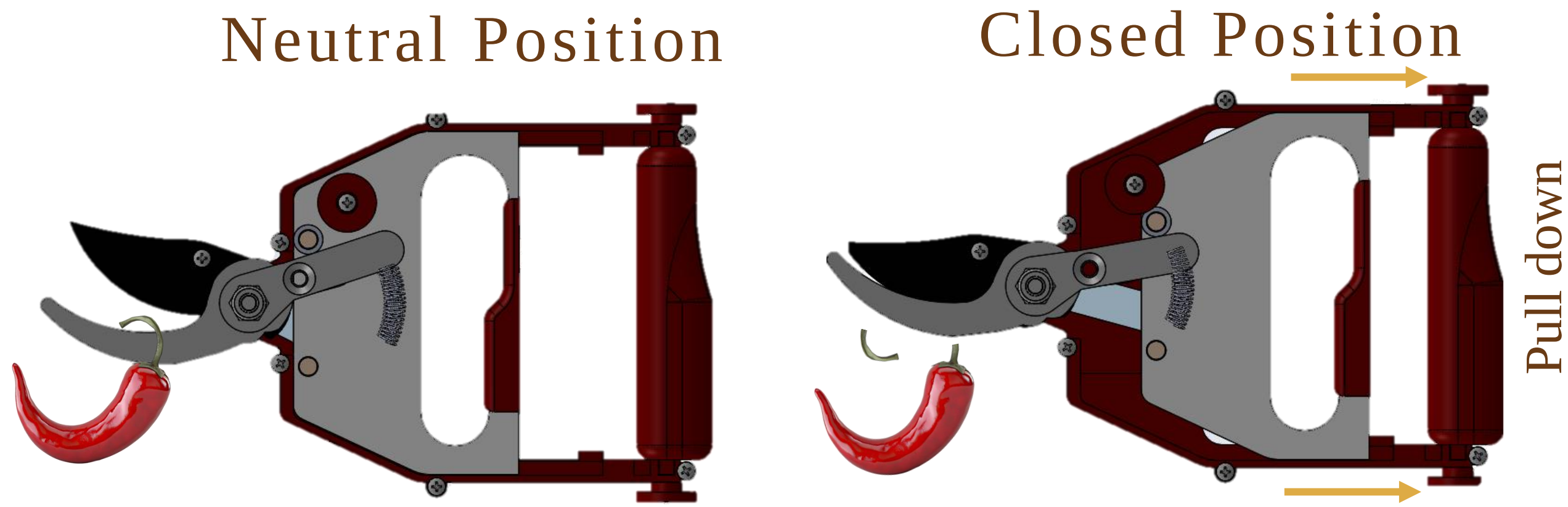
- Pulling over squeezing
- Straight vs bent wrist
- Lighter
- Ergonomic handles

Robustness

- Durable material, ABS
- Replaceable blades



How it Works



How is it Better?

- 0.6oz lighter
- Straight wrist orientation
- Applies stress away from sensitive parts of the hand
 - Utilizes the stronger hand muscles
- Similar required effort as market products
- Locking mechanism and actuators do not rust
 - 19% increase in max repetitions

Hand Fatigue Test

Recorded maximum actuations on market shear and redesigned shear before feeling “fatigued.” Averaged 19% increase in actuation.

Max reps with Market Shear	Max reps with GRIP shear	% change in rep number
234	246	5.13%
110	125	13.36%
176	199	13.07%
56	78	39.29%
160	203	26.88%

User Feedback

Smile Farms Clients

- Questioned about the comfort and intuitive nature of the design
- Found our product a lot more comfortable to hold
- Could see themselves using our product for a long time

Applications

Clients can be expanded to people with hand mobility issues such as arthritis and the elderly that are struggling to continue gardening.

Acknowledgements

We would like to thank our advisor, Professor David Wootton, and the shop staff for our design consultations. We would also like to thank NYSID for sponsoring this project, and CP Unlimited, Jessica Ann Francese, and the Smile Farms employees for their support and feedback.

Next Steps

We want to improve tolerances to have a smoother motion, add a softer and more comfortable grip, increase mechanical advantage, and distribute the weight of shear more evenly.